6 A simple ideal iron-cored transformer is illustrated in Fig. 6.1.
An alternating input p.d. V_p of 240 V is connected across the primary coil, of 120 turns of the transformer. The secondary p.d. V_s of 3600 V is applied to a load resistor.

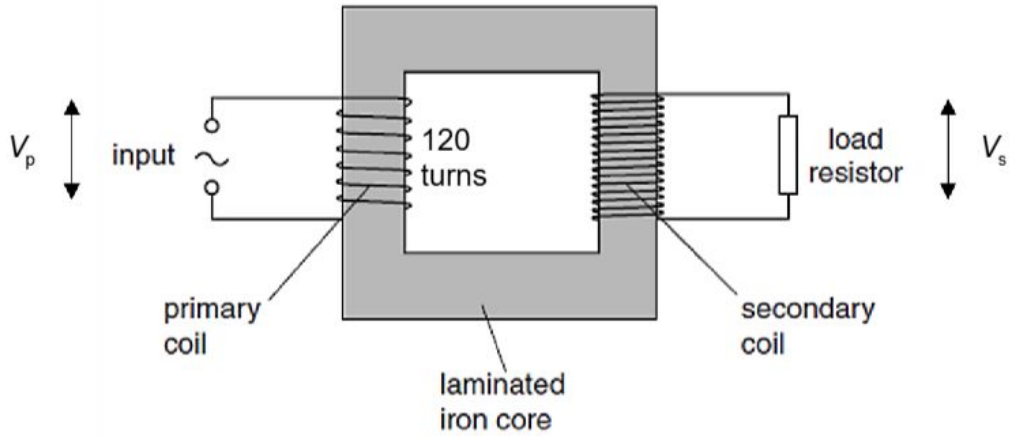


Fig. 6.1

(a) **(i)** Explain why an iron core is preferred over an air core in the construction of a transformer.

. it increases the magnetic flux density OR concentrates the magnetic flux as it links it from the primary to the secondary coil, (thereby ....
. increase the efficiency of energy transfer) [B1]

..... [1]

(ii) Explain why the iron core is laminated.

. to reduce energy losses by reducing eddy (induced) currents [B1] .....

.....

..... [1]

(b) Calculate the number of turns in the secondary coil.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \Rightarrow N_s = 120\left(\frac{3600}{240}\right) = 1800 \quad \text{[B1]}$$

number of turns = ... [1]

(c) Using Faraday's Law, explain why an alternating current in the primary coil gives rise to an alternating e.m.f. in the secondary coil.

The alternating current in the primary coil produces a **magnetic flux that changes continuously** in both magnitude and direction. [B1]

Since the secondary coil is wound on the same iron core, **this changing magnetic flux links the secondary coil**, so the flux through the secondary coil also alternates. [B1]

By Faraday's Law, the induced e.m.f. is proportional to the rate of change of magnetic flux linkage [$\varepsilon = -N(d\Phi/dt)$]. Since the flux is continuously changing, an e.m.f. is continuously induced in the secondary coil. **As the flux alternates, the rate of change also alternates in sign,** so the induced e.m.f. alternates as well. [B1]

..... [3]

(d) A step-up transformer is used when transmitting electricity over long distances.

Suggest a reason for this.

A step-up transformer increases the voltage and reduces the current, thereby reducing the power loss in the transmission cables ($P = I^2R$), making transmission more efficient. [B1]

..... [1]

[Total: 7]